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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Flow cytometry data were collected by FlowJo software v10 (BD); ELISA and Neu50 titers were analyzed by Excel 2013 (Microsoft); Measurement of the size of circRNA-LNPs was performed by Zetasizer software (version 7.13); qRT-PCR data were collected and analyzed by CFX Manager 3.1 (Bio-Rad); Images of brain tissue sections were captured and analyzed by TissueFAXSViewer_7.1.7.123 (TissueGnostics).
Data analysis	Statistical analysis were computed by GraphPad Prism 8 (GraphPad); Figures were created using Adobe Illustrator 2021 (Adobe Systems Inc.) or PowerPoint 2013 (Microsoft).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The data supporting the findings of this study are available within the paper (and its supplementary information files). Source data are provided with this paper.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	No human research was performed
Reporting on race, ethnicity, or other socially relevant groupings	No human research was performed
Population characteristics	No human research was performed
Recruitment	No human research was performed
Ethics oversight	No human research was performed

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample size of mouse experiments were determined based on the mean and standard derivations observed in our previous vaccine and antibody studies in mouse models (Liu X et al., NPJ Vaccines 2018; Yu L et al., mBio 2021). For in vitro experiments, no sample size calculation was performed. All the in vitro and in vivo experiments were repeated successfully for at least two times.
Data exclusions	All data is included and shown.
Replication	Titers of binding and neutralizing antibodies were always measured in two technical replicates. Genome copy numbers of ZIKV or DENV in either the sera or the tissues were measured in three technical replicates. All the experiments regarding mice models were successfully repeated for at least two times.
Randomization	Mice (either C57BL/6 or Ifnar-/- or neonatal ICR mice) were randomly allocated into groups.
Blinding	The experiments involving virological and immunological analysis of mice samples were performed by technicians, who were blinded to the identities of groups. For the histopathological analysis of tissue sections, a single-blind manner was employed. As for the rest in vitro experiments that did not involve mice samples, blinding was not essential to avoid any kind of bias.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

-Western blot analysis
anti-ZIKV E mAb, clone: 8D10, produced and kept in-house;
anti-ZIKV NS1 mAb, clone: 749-A4, produced and kept in-house;
mouse anti-GAPDH antibody (Beyotime, AF5009) ;
HRP-conjugated goat anti-human IgG antibody (Beyotime, A0201);
HRP-conjugated goat anti-mouse IgG antibody (Beyotime, A0216).

-ELISA
HRP-conjugated goat anti-mouse IgG antibody (Beyotime, A0216).

-FNT
Alexa Fluor 647-conjugated goat anti-human IgG antibody (SouthernBiotech, 2040-31).

-in vitro ADE
DENV E cross-reactive mAb, clone: ZK8-4, produced and kept in-house;
Alexa Fluor 647-conjugated goat anti-human IgG antibody (SouthernBiotech, 2040-31).

-Flow cytometry
CD3 (PerCP/Cy5.5, clone: 17A2, BioLegend);
CD4 (APC, clone: RM4-5, BD);
CXCR5 (PE, clone: 2G8, BD);
PD-1 (BV421, clone: EH12.1, BD);
CD45R (PE/Cy7, clone: RA3-6B2, BD);
CD38 (PerCP/Cy5.5, clone: 90/CD38, BD);
CD138 (PE, clone: 281-2; BD);
CD95 (BV421, clone: Jo2, BD);
GL7 (Alexa Fluor 647, clone: GL7, BD).

-ICS
mAb α-CD16/32;
CD3 (BV421, clone: 17A2, BioLegend);
CD4 (APC, clone: RM4-5, BD);
CD8α (FITC, clone: 53-6.7, BioLegend);
IFNγ (PE, XMG1.2, BD);
TNFα (APC/Cy7, clone: MP6-XT22, BioLegend);
IL-2 (BV605, clone: JES6-5H4, BioLegend).

Validation

anti-ZIKV E mAb, clone: 8D10, was validated in Liu X. et al. , NPJ vaccines 2018;
anti-ZIKV NS1 mAb, clone: 749-A4, was validated in Wessel AW et al., Nature communications 2020;
DENV E cross-reactive mAb, clone: ZK8-4, was validated in Lei Y et al., JCI insight 2017;
The rest antibodies are commercially available. Their respective validation information is available on the manufacturers' websites.

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)	African green monkey kidney Vero cells (CCL-81), human embryonic kidney 293T cells (CRL-3216), human lung cancer A549 cells (CCL-185), and human chronic myelogenous leukemia K562 cells (CCL-243) were purchased from American Type Culture Collection (ATCC); Expi293F cells were purchased from Thermo Fisher Scientific (R79007).
Authentication	Authentication was performed by the suppliers.
Mycoplasma contamination	The cell lines were tested by the suppliers and there is no mycoplasma contamination. All of them were tested monthly once

	received in-house, and all were tested negative for mycoplasma.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used in this study.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	C57BL/6 mice, 8 weeks old at the beginning of study; neonatal C57BL/6 mice, 1 day old at the beginning of study; Ifnar-/- C57BL/6 mice, 12 weeks old at the beginning of study; CD-1(ICR) mice, 1 day old at the beginning of study.
Wild animals	This study did not involve wild animals.
Reporting on sex	This study employed three mice models. As for the maternal immunization and neonatal challenge model, the immunization sassy was performed using female C57BL/6 mice, but the challenge assay involved both female and male pups. As for the adult Ifnar-/- mice model, we used both male and female mice. As for the challenge assay regarding mouse-adapted DENV2, we used both female and male 1-day-old ICR mice. Therefore, the mainly findings of this study are not limited to one particular sex. Our primary objective focused on the general immunogenicity and protective effects of the vaccine across the population, rather than investigating sex-specific responses. Consequently, although we included both sexes, detailed analysis to determine potential sex-based differences in immune response were not conducted.
Field-collected samples	This study did not involve field-collected samples.
Ethics oversight	All animals were bred and housed in the Animal Experimental Center of Guangzhou Institutes of Biomedicine and Health (GIBH), Chinese Academy of Sciences (CAS). All experiment protocols were approved by the Institutional Animal Care and Use Committee of GIBH (IACUC, No. 2021061). Studies involving ZIKV or DENV challenge assays were conducted under animal biosafety level 2 (BSL-2) plus conditions in the Animal Experimental Center of GIBH, CAS.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	Not applicable
Novel plant genotypes	Not applicable
Authentication	Not applicable

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	For the analysis of GC reaction, ILN lymphocytes were isolated by passing through 200-mesh stainless steel wire meshes. Cells were blocked with an Fc receptor antibody α-CD16/32 (clone: 2.4G2) on ice for 5 min, followed by staining with fluorochrome-conjugated antibodies in PBS with 2% FBS: for the Tfh cell panel, CD3 (PerCP/Cy5.5, clone: 17A2, BioLegend), CD4 (APC, clone: RM4-5, BD), CXCR5 (PE, clone: 2G8, BD), PD-1 (BV421, clone: EH12.1, BD); for the GC B cell and plasma cell panels, CD45R (PE/Cy7, clone: RA3-6B2, BD), CD38 (PerCP/Cy5.5, clone: 90/CD38, BD), CD138 (PE, clone: 281-2; BD), CD95 (BV421, clone: Jo2, BD), GL7 (Alexa Fluor 647, clone: GL7, BD). The staining was performed on ice for 30 min. Cells were fixed and analyzed by LSRFortessa (BD). For the ICS assay, lymphocytes were isolated from the spleens by passing through 200-mesh stainless steel wire meshes followed by density gradient centrifugation using mouse lymphoid separation medium (Dakewe). Cells were seeded into 96-

well plates (2 × 10^6 cells per well) and stimulated with overlapping peptide pools (2 μg/ml of each peptide) derived from ZIKV E. After incubated at 37 °C for 2 h, cell cultures were added with brefeldin A (10 mg/ml; BD) and incubated at 37 °C for another 10 h. After washed twice with PBS, cells were blocked with mAb α-CD16/32 on ice for 5 min and stained with the following surface marker antibodies in PBS with 2% FBS on ice for 30 min: CD3 (BV421, clone: 17A2, BioLegend), CD4 (APC, clone: RM4-5, BD), and CD8α (FITC, clone: 53-6.7, BioLegend). Cells were then fixed and permeabilized with Cytofix/Cytoperm buffer at 4 °C for 30 min. Finally, cells were stained with intracellular antibodies against IFNγ (PE, XMG1.2, BD), TNFα (APC/Cy7, clone: MP6-XT22, BioLegend) and IL-2 (BV605, clone: JES6-5H4, BioLegend) and analyzed on LSRFortessa.

For the FNT and in vitro ADE assays, Vero cells or K562 cells were fixed and permeabilized using Cytofix/Cytoperm buffer, labeled with mAb 8D10 (anti-ZIKV E) or mAb ZK8-4 (cross-reactive with DENV E proteins), and stained with Alexa Fluor 647-conjugated goat anti-human IgG antibody. Cells were analyzed on Accuri C6 flow cytometer (BD).

Instrument: LSRFortessa and Accuri C6 flow cytometer (BD)

Software: FlowJo software v10 (BD)

Cell population abundance: N/A

Gating strategy: FSC-A vs FSC-H and SSC-H vs SSC-W parameters were used to exclude doublets. For the ILN lymphocytes, we firstly gated on CD3+CD4+ cells. Within this subset, we identified PD-1+CXCR5+ cells as the Tfh population. We also identified CD45R+CD138 + population as the plasma cells. Meanwhile, we gated on CD45R+CD38- cells, and within this subset we identified CD95+GL7 + GC B cells.

For the splenic lymphocytes, we gated on two subsets: CD3+CD4+ subsets and CD3+CD8+ subsets. Subsequently, we identified the IFN-γ+, TNF-a+, or IL-2+ subpopulation from these two subsets.

For the FNT and in vitro ADE assay, FSC-A vs FSC-H and SSC-H vs SSC-W parameters were used to exclude doublets. Next, we gated on the ZIKV E+ cells or DENV E+ cells.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.